

# FORREST WILLIAMS

Email: [forrestfwilliams@icloud.com](mailto:forrestfwilliams@icloud.com)

GitHub: <https://github.com/forrestfwilliams>

Ankeny, Iowa

## PROFESSIONAL EXPERIENCE

|                                                                                                                                                                                                                                                                  |                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| <b>Senior Radar Systems Engineer, Ursa Space Systems:</b> Remote                                                                                                                                                                                                 | 2025 to Present |
| <ul style="list-style-type: none"><li>- Leading an edge-computing airborne moving target ID (AMTI) project</li><li>- Developing a Claude Skill for SAR intelligence</li></ul>                                                                                    |                 |
| <b>Senior Research Software Engineer, Alaska Satellite Facility:</b> Remote                                                                                                                                                                                      | 2022 to 2025    |
| <ul style="list-style-type: none"><li>- Developed RTC capabilities for commercial SAR data in the SICD format</li><li>- Lead the <a href="#">ASF Displacement Portal</a> visualization project</li><li>- Lead the Sentinel-1 Burst data access project</li></ul> |                 |
| <b>PhD, Massey University:</b> Palmerston North, New Zealand                                                                                                                                                                                                     | 2019 to 2022    |
| <ul style="list-style-type: none"><li>- Developed Sentinel-1 InSAR and optical data workflows to monitor landslides across New Zealand</li><li>- Developed MintPy utilities to ingest HyP3 InSAR products</li></ul>                                              |                 |
| <b>NASA Jet Propulsion Laboratory Intern:</b> Remote                                                                                                                                                                                                             | 2021            |
| <ul style="list-style-type: none"><li>- Created a Python-based UAVSAR workflow to monitor earthquakes and landslides via high-resolution InSAR analyses</li></ul>                                                                                                |                 |
| <b>MS, Iowa State University:</b> Ames, Iowa USA                                                                                                                                                                                                                 | 2017 to 2019    |
| <ul style="list-style-type: none"><li>- Built an ArcPython toolbox leveraging high-resolution imagery and terrain data to track river movements</li><li>- Performed continuous in-field discharge monitoring for a flood-prone stream</li></ul>                  |                 |

## SOFTWARE ENGINEERING PROJECTS AND CAPABILITIES

|                                                                                                                 |  |
|-----------------------------------------------------------------------------------------------------------------|--|
| <b>Lead Developer:</b> <a href="#">MultiRTC</a> Geocoding and RTC processing for commercial and public SAR data |  |
| <a href="#">SatChip</a> AI data preparation for public satellite missions                                       |  |
| <a href="#">Burst2Safe</a> Custom-size Sentinel-1 SLC file generation                                           |  |
| <a href="#">INDEX_SAFE</a> Sentinel-1 burst extraction library                                                  |  |
| <b>Contributor:</b> MintPy, HyP3, HyP3-ISCE2, HyP3-GAMMA, AutoRIFT, ISCE2                                       |  |
| <b>Analysis Types:</b> AMTI, MTI, Time-Series InSAR, Change Detection, Terrain Analysis                         |  |
| <b>Key Technologies:</b> Python, Claude, C++, Cython, Pybind11, GDAL, STAC, AWS, PyTorch, QGIS                  |  |

## EDUCATION

|            |                                                                                                                                                          |      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>PhD</b> | Massey University: Palmerston North, New Zealand<br>Interned with the Jet Propulsion Laboratory's SAR group<br>Major: Earth Science, GPA: Not applicable | 2022 |
| <b>MS</b>  | Iowa State University: Ames, Iowa USA<br>Graduate Certificate in GIS<br>Major: Environmental Science, GPA: 3.77                                          | 2019 |
| <b>BA</b>  | Carleton College: Northfield, Minnesota USA<br>Major: Geology, GPA: 3.54                                                                                 | 2016 |